MIL-M-14946 (MU)
17 June 1971

MILITARY SPECIFICATION

MINE, WARNING, ANTI-INTRUSION, PRACTICE XM46

LOADING AND ASSEMBLING FOR

1. SCOPE

1.1 This specification covers the loading and assembling of one type of mine designated as Mine, Warning, Anti-Intrusion, Practice XM46.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-B-14950	-	Bags and Housings for Aerial Mines

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FEDERAL

FED-STD No. 751	-	Stitches, Seams, and Stitching

FSC: 1345

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DRAWINGS

U.S. ARMY MUNITIONS COMMAND

9223687 - Mine, Warning, Anti-Intrusion, Practice XM46

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Material

3.1.1 General.-Material and parts shall be in accordance with applicable drawings and specifications.

3.1.2 Friction plate.-The friction plate shall be fabricated in accordance with the applicable drawing and specifications.

3.2 Mine.-The mine, warning, anti-intrusion, practice XM46 shall comply with requirements specified on Drawing (dwg.) 9223687 and with all requirements specified in applicable specifications.

3.3 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of tears, rips, dirt, dust, and other foreign matter. Stitches, seams, and stitching shall be made in accordance with FED-STD No. 751. Stitches shall not be loose, poorly formed, crowded, tight, crooked, or skipped. Seams and stitching shall be free of puckers, twists, pleats, undulations, and raised seams, and shall not have raw edges exposed (felled seams).

3.4 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article sample by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth

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in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of chemical analysis and certificates of conformance on all materials purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents. The sample size of chemical materials analyzed and results of the analysis shall be recorded on the above-mentioned certificate of analysis (see 6.4).
f. Number of items in the lot.
g. Date submitted.

The certificates shall be signed by a responsible agent of the certifying organization. The initial certificates submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-Prior to the start of normal production, the contractor shall submit a first article sample as designated by the Contracting Officer, for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of one hundred (100) mines, and fifteen (15) complete sets of parts (i.e., 15 each of every component and every sub-assembly) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production process, procedures and equipment which will be used in fulfilling the contract. All parts and material, including packaging and packing shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and

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submit a record of this inspection with the sample, including certificates of conformance and analysis for materials (see 4.1.1.e). A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections and tests to be performed.-Each assembly shall be inspected and tested by the Government for all the inspections specified in 4.3.2 of this specification, for all the requirements of the applicable drawing, and for all tests indicated herein.

4.2.2.1 Quantities for inspection test.-After classification of defects and other inspections have been performed as specified in this specification, the following quantities shall be subjected to the indicated tests.

4.2.3 Rejection.-If any mine fails to comply with the applicable requirements, as listed in 4.2.2, the first article sample shall be rejected. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit new or supplemental quantities until such time as an acceptable sample is submitted, or until otherwise directed by the Contracting Officer.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification, or drawing not affecting safety, performance, interchangeability or storage, as determined by the Government inspector, shall not be deemd to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105 and shall be numbered in accordance with MIL-STD-1168. The end item lot size shall be not less than the number of units (plus samples) which are packaged in one canister; neither shall the end item lot size be greater than two hundred thousand (200,000) units, or one week's production - whichever comes first - unless otherwise specified by the Contracting Officer. Inspection lots for components or sub-assemblies, other than the items of delivery, shall be homogeneous and of a size convenient to both the contractor and the Government inspector.

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4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105 except that inspection for critical defects shall be 100 percent. Continuous sampling plans in accordance with MIL-STD-1235 may be used, for defects other than critical, if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed, using an AQL of 0.40 percent for each Major defect and an AQL of 0.65 percent for each Minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Initiator Loading Assembly, (see dwg. 9255405)

Categories	Defects	Method of Inspection	Defect Code
Critical:	None Defined		
Major:	None defined		
Minor:			
201.	Sand missing	Visual	03001
202.	Evidence of poor workmanship (see 3.4)	Visual	03002

4.3.2.2 Mine, Warning, Anti-Intrusion, Practice XM46 (See dwg. 9223687)

Categories	Defects	Method of Inspection	Defect Code
Critical:	None defined		
Major:	None defined		
Minor:			
201.	Missing or damaged iniator loading assembly	Visual	05001
202.	Missing friction plate	Visual	05002
203.	Contents exposed	Visual	05003
204.	Assembly damaged	Visual	05004
205.	Evidence of poor workmanship	Visual	05005

4.3.3 Testing

4.3.3.1 Dimensional control

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4.3.3.1.1 Dimensional control of punched sheet parts.-In place of the normal sampling associated with the classification of defects, and with the approval of the Contracting Officer, a sample of at least three (3) parts may be dimensionally inspected to qualify the tools used in the forming process for use in production. In addition, a random sample of three (3) parts shall be selected after the production of each 5,000 parts or at the end of each shift, provided the number of parts produced during that shift is less than 5,000, for dimensional inspection as a control of the tools during production. Of the three (3) samples selected, one (1) shall be the last part produced.

If any defective parts are found during qualification of the tools, the tools producing the defective parts shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective parts shall be removed from production, and that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.3.3.1.2 Dimensional control of molded parts.-In place of the normal sampling associated with the classification of defects, and with the approval of the Contracting Officer, and after a curing time for the parts has been approved (see Note 1), a sample of three (3) parts (as molded) from each cavity may be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of each cavity during production.

The inspection of the parts shall be performed only when the curing time for the parts has been reached. The inspection for control of each cavity during production shall be performed on the above quantities from each cavity after production of each 25,000 parts or at the end of each shift, providing the number of parts produced during that shift is less than 25,000 parts. Of the three (3) samples, one (1) sample shall be the last part produced. Parts which fail to comply with dimensional requirements after dimensional stability has been attained shall be classed defective.

All cavities formerly removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to substantiate the request to the Contracting Officer.

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(NOTE 1) - In establishing a curing time, dimensionally check ten (10) parts from each cavity at fifteen (15) minute intervals until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to Picatinny Arsenal, SMUPA-ND8. If any new change in material, cycle time, a reworked cavity, or new cavity is used, a new curing time shall be established and approved.

4.3.4 Inspection equipment.- Equipment Tabulation (ET) number and Equipment Lists (EL's) referenced theron identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation.- Inspection equipment drawings and lists provided and revised in accordance with the requipements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.-The manufactured mines shall be loaded into the canister of the delivery system in accordance with the applicable specification. Interim storage shall be in accordance with 6.5.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (see 6.3.5).

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6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.- The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army Designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (see 6.3.4) or commercial equipment selections must be approved by the Government prior to fabrication or procurement of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.- Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls:

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All submitted contractor designs shall conform to Specification MIL-D-1000 Category A, Form 3. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor design.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, Attn: SMUPA-ND13. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Submission of test data, certificates of chemical analysis, and ammunition data cards.-In addition to the normal distribution of records for loaded items procured by the Department of the Army, one copy of ammunition data cards shall be forwarded to the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey 07801
Attn: SMUPA-ND8

6.5 Interim storage

6.5.1 Definition.-Interim storage is considered to be any delay after final assembly, prior to packing the mines as required by the contract.

6.5.2 Conditions.-When interim storage is utilized, the mines shall be handled, stored, and monitored in a safe and approved manner. Particular attention shall be given to item density which should not be more than one (1) mine per 0.5 cubic inches. In addition, the height of the packing level shall not exceed the diameter of the container, and the method and weight used to keep the mines submerged shall not cause the mines to be crushed or the packing density to change in storage. Attention shall also be given to sufficient quantity of desensitizer, condition and material of the containers, and the evaporation rate of the desensitizer for the particular storage area and container. Failure to maintain the assemblies in their accepted condition will result in re-inspection and re-test as directed by the Government.

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6.6 Submission of results of contractor-conducted exaiminations and tests.-Unless otherwise specified by the contracting officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, Attn: SMUPA-ND8.

Custodian:
Army-MU

Preparing Activity:
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